



The influence of biotinylation on the ability of a computer designed protein to detect B-cells producing anti-HIV-1 2F5 antibodies

Danilo F. Coêlho ^{a, b}, Matheus V.F. Ferraz ^{a, b}, Ernesto T.A. Marques ^{a, c}, Roberto D. Lins ^{a, b, *}, Isabelle F.T. Viana ^{a, **}

^a Aggeu Magalhães Institute, Oswaldo Cruz Foundation, Recife, PE, 50670-465, Brazil

^b Department of Fundamental Chemistry, Federal University of Pernambuco, Recife, PE, 50740-540, Brazil

^c Department of Infectious Diseases and Microbiology, University of Pittsburgh, Pittsburgh, PA, 15261, USA

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ABSTRACT

Antibodies against the HIV-1 2F5 epitope are known as one of the most powerful and broadly protective anti-HIV antibodies. Therefore, vaccine strategies that include the 2F5 epitope in their formulation require a robust method to detect specific anti-2F5 antibody production by B cells. Towards this goal, we have biotinylated a previously reported computer-designed protein carrying the HIV-1 2F5 epitope aiming the further development of a platform to detect human B-cells expressing anti-2F5 antibodies through flow cytometry. Biophysical and immunological properties of our devised protein were characterized by computer simulation and experimental methods. Biotinylation did not affect folding and improved protein stability and solubility. The biotinylated protein exhibited similar binding affinity trends compared to its unbiotinylated counterpart and was recognized by anti-HIV-1 2F5 antibodies expressed on the surface of patient-derived peripheral blood mononuclear cells. Moreover, we present a high affinity marker for the identification of epitope-specific B cells that can be used to measure the efficacy of vaccine strategies based on the HIV-1 envelope protein.

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1. Introduction

Antibodies are the most common markers used to detect and quantify binding and recognition of specific antigens. In natural infections, their production is typically elicited by pathogen proteins and/or carbohydrates. Prophylactic strategies often rely on the use of antigens (e.g., viral particles, attenuated viruses, fusion protein domains) to mimic the host immune response upon pathogen exposure. Nevertheless, conventional approaches have faltered for several viruses, such as the Human Immunodeficiency virus – HIV. This virus has evolved an arsenal of molecular tricks to avoid or mitigate immune responses [1], including the transient exposure of key epitopes capable of triggering the production of broadly neutralizing antibodies. One of these cryptic epitopes is termed 2F5 epitope, whose core is a highly conserved continuous

9-mer amino acid sequence in the ectodomain of the gp41 envelope protein from HIV, also known as membrane-proximal external region (MPER) [2]. While the antibodies raised by this epitope (anti-2F5 antibodies) are well known to be produced by only a minority of infected individuals after 2–3 years of exposure [3], these antibodies are one of the most powerful and broadly protective anti-HIV antibodies described so far. They can neutralize over than 50% of viral isolated panels and protect primate models upon challenge [4,5]. Accordingly, the protective potential of a vaccine strategy can be assessed by measuring its capability of eliciting anti-2F5 antibodies upon immunization.

The identification of anti-2F5 antibodies *in vitro* has been a difficult task since their detection relies on the recognition of the MPER peptide. When free in solution, this peptide is poorly recognized by human antibodies, which hampers the assessment of the protective value of the vaccines under development. To overcome this limitation, we have previously engineered a chimerical protein where a 9-mer (ELDKWASLW), based on the 2F5 epitope, was grafted onto Top7, a computationally devised protein [6]. The protein named Top7-2F5 was shown to be specifically recognized by the respective monoclonal antibody (mAb 2F5) by means of

* Corresponding author. Aggeu Magalhães Institute, Oswaldo Cruz Foundation, Recife, PE, 50670-465, Brazil.

** Corresponding author.

E-mail addresses: roberto.lins@cpqam.fiocruz.br (R.D. Lins), isabelle.viana@cpqam.fiocruz.br (I.F.T. Viana).

enzyme-linked immunosorbent assay (ELISA) [7,8]. In this work, we assessed the influence of the biotinylation of the Top7-2F5 aiming the development of a system for *in vitro* detection of patient-derived B cells expressing the anti-2F5 antibody. This platform holds the potential to help measuring the protective efficacy of vaccines on clinical trials.

Protein conjugation with biotin has been long used for a plethora of techniques, such as chromatographic protein purification, chemical precipitation, immunohistochemistry, drug delivery and immunoassays [9–14]. The identification of a biotin-labeled biomolecule occurs via specific binding to avidin/streptavidin and the formation of a highly stable complex [15–18]. Characterization of the chemical bond between the biotin molecule and the target protein has also been extensively studied. Published works suggest that the bond stability indeed depends on the nature of the chemical bond [19–22].

Previous studies on protein biotinylation have shown that the main advantage of using this labeling technique is that physical properties and biological activity of the biotinylated protein are only minimally affected. Nevertheless, it is not always the case and therefore the biotin-labeled protein should be characterized prior to its use [23]. For instance, while biotin conjugation of antibodies may lead to a reduced antigen-binding capacity [14,24], biotinylation of protein A was found to prevent IgM antibody binding and decrease the sensibility of an ELISA assay [12]. In most cases, only the changes in protein behavior upon biotinylation are discussed, leaving the molecular basis underlying these changes unknown. Only few studies have investigated possible structural changes due to protein biotinylation. Azim-Zadeh et al. have shown that conjugation with biotin did not induce structural changes on bovine serum albumin (BSA), when assessed by circular dichroism (CD) [25]. The BSA molecule forms 17 intramolecular disulfide bonds, and the structural compactness of BSA is the reason why this protein may not undergo conformational changes upon biotinylation. Far- and near-UV CD spectroscopy have also been used to monitor the secondary structure change on the biotinylated human lysozyme (BioHuL) [26]. This study has shown that BioHuL exhibits secondary structure content and thermal stability similar to its non-biotinylated counterpart. Although remarkable changes in protein structure were not observed, biotinylation led to a decrease of biological function of BioHuL, which retained only 60% of its enzymatic activity when compared to the native lysozyme.

In this work, the Top7-2F5 protein was covalently linked to a biotin molecule and the specific binding to the B cell receptor (BCR) was detected by directly measuring the resulting fluorescence signal by flow cytometry. As the biotinylation takes place in primary amines, this is even more worrisome in the case of the 2F5 epitope, since a lysine residue is found in the middle of the epitope. Therefore, possible negative effects on stability and immunoreactivity of the Top7-2F5 protein must be assessed. In the present work we have characterized the structural stability of biotinylated Top7-2F5 protein, its immunoreactivity against the 2F5 mAb and its ability to be recognized by and to identify B cells expressing the anti-2F5 antibody *in vitro*.

2. Experimental and methods

2.1. Molecular dynamics (MD) simulations

Initial coordinates for the native Top7 protein were taken from PDB ID: 1QYS. For the construction of Top7-2F5, the residues T22 and T23 of Top7 were removed and the 2F5 epitope sequence in a loop conformation was added into the scaffold [7]. The system modeling was performed using the Swiss-PDB Viewer 4.0.1 software [27]. The biotin molecular topology was built based on

parameters retrieved from GROMOS53A6 force field (Fig. S1 and Table S1) [28]. A short MD simulation of 15 ns was performed and the molecular geometry of the biotin rings was shown as boat conformation, according to experimental X-ray diffraction data [29] (data not shown for conciseness). The biotinylated versions of Top7 and Top7-2F5 were built by covalently linking it to all available primary amines to the biotin carboxyl groups. The modeling of biotinylated proteins has been made assuming full biotinylation of all free amino residues, since the biotinylation process takes place with excess of biotin reagent. All structures were centered into a cubic box of 10.0 × 10.0 × 10.0 nm dimensions and explicitly solvated using SPC water model [30]. Periodic boundary conditions were used in the x, y, and z directions. The LINCS method was used to constrain all bonds involving hydrogen atoms [31]. Solute and solvent temperatures were separately coupled to a thermostat using the velocity rescale scheme [32] with three different reference temperatures: 300 K, 340 K and 380 K and a relaxation time of 0.2 ps. The pressure was maintained at 1 bar by isotropic coordinate scaling using the Berendsen barostat [33] with a relaxation time of 0.2 ps and compressibility (κ_T) of $4.5 \times 10^{-5} \text{ bar}^{-1}$. A short-range cutoff radius of 1.4 nm was used for all non-bonded interactions. Long-range electrostatic interactions were taken into account using the reaction field method [34] with $\epsilon = 66$ beyond the cutoff radius of 1.4 nm. Counter ions were added to ensure a neutral system net charge and a salt concentration of 0.150 mM. The systems were initially energy optimized using 5,000 steps of the steepest descent algorithm. Integration was carried out using an integration time step of 2 fs based on the leapfrog algorithm [35]. Thermalization of all systems was carried out in NVT ensemble in the respective temperature for 20 ps followed by a pressure equilibration in a NPT ensemble for 1 ns. All equilibration steps were performed with position constraints for all atoms. MD simulations were carried out using the NPT ensemble and performed for 100 ns using the GROMOS 53A6 force field [28] with GPU acceleration and the Verlet cut-off scheme [36]. All simulations and analyses were performed using the GROMACS 4.6.5 simulation package [37], DSSP (Database of Secondary Structure in Proteins) software [38] and visual analyses were carried out using the Visual Molecular Dynamics software (VMD) 1.8.7. Representative structures of the biotinylated and non-biotinylated versions of Top7 and Top7-2F5 proteins were selected from the trajectory upon system equilibration (last 50 ns) by clustering all of the frames within this interval employing g_cluster tool from GROMACS suite and aligning the backbone atoms with a 0.14 nm cutoff. Each single conformational cluster was used to perform electrostatic surface potential analyses using the MEAD v2.2 software [39].

2.2. Recombinant protein expression and purification

The chimeric proteins were produced in prokaryotic system and purified by affinity chromatography as previously described [7]. Briefly, the DNA sequences coding for Top7 and Top7-2F5 proteins were flanked by two enzyme restriction sites (5' Hind III and Nhe I and 3' Nco I and Xho I) and codon optimized for translation in bacterial system. The optimized genes were synthesized by a commercial supplier (GeneScript) and inserted in the pRSET A expression vector (Invitrogen) for expression as N-terminally histidine-tagged proteins. Bacterial expression of the proteins was carried out in the *Escherichia coli* BL21 Star (DE3)pLysS (Invitrogen) strain harboring the expression vectors. Cells were grown in Luria-Bertani – LB medium supplemented with 100 mg/mL of ampicillin (LB-AMP) to an OD₆₀₀ of 0.5 and protein expression was induced with 1 mM isopropyl β-D-1-thiogalactopyranoside – IPTG at 37 °C, 4 h, 225 rotations per minute – rpm. Following, expressing bacterial pellets were harvested by centrifugation at 4 °C, 30 min,

5000×g and resuspended in lysis buffer (300 mM NaCl, 50 mM Tris-HCl, 20 mM Imidazole, pH 8.0) supplemented with 1X protease inhibitor cocktail (Roche). Bacterial cells were disrupted by sonication on ice in a Vibracell VCX 750 Sonicator (amplitude set at 40%) by 6 pulses of 30 s at 60 s intervals. The lysate was clarified by centrifugation at 4 °C, 30 min, 8000×g and the recombinant proteins were purified through affinity chromatography with a nickel matrix column (Ni-NTA – QIAGEN) (Fig. S2). Protein concentration was determined by spectrophotometry and corrected according to the respective molar extinction coefficients. The Top7-2F5 sequence data have been submitted to the GenBank databases under the accession number MK205154. The Top7 sequence data can be retrieved from PDB ID 1QYS.

2.3. Protein biotinylation

Prior to biotinylation, the recombinant proteins were dialyzed against 1X Phosphate Buffered Saline – PBS, to remove any traces of primary amines (e.g., Tris), which could decrease the protein biotinylation efficiency. The biotinylation procedure was carried out using the Thermo Scientific Pierce EZ-Link Sulfo-NHS-Biotin kit, following the manufacturer's instructions. Briefly, a fresh 10 mM solution of the biotin reagent was prepared and added to each protein solution targeting a 20-fold molar excess of biotin reagent for labeling. The biotinylation reactions took place for 2 h on ice. Following, the labeled proteins were purified for optimal performance and stability by removing the excess non-reacted and hydrolyzed biotin reagent by dialysis to a buffer containing 300 mM NaCl, 50 mM Tris-HCl, pH 8.0.

2.4. Enzyme-linked immunosorbent assay (ELISA)

High binding, half area 96-well polystyrene plates (Costar; Lowell, MA, USA) were coated with 32 pmol of each protein (Top7 and Top7-2F5) or peptide (MPER) per well for 16 h at 4 °C. The reactions were blocked with 5% (w/v) skimmed milk (Bio-Rad) in PBS-T buffer (1X PBS with 0.05% (v/v) Tween-20) for 15 min at room temperature. Following, the monoclonal antibody directed to the 2F5 epitope (Polymun Scientific) was diluted from 1 ng mL⁻¹ to 1 × 10⁻¹⁰ ng mL⁻¹ in assay buffer (5% (w/v) skimmed milk in PBS-T) and added to the plate for 2 h at room temperature. Plates were washed five times with PBS-T to remove any unbound antibody. Following, peroxidase-conjugated goat anti-human IgG antibody (Jackson ImmuneResearch) was diluted to 1:35,000 in assay buffer and added to the wells. Plates were incubated for 1 h at room temperature, washed four times with PBS-T and allowed to soak in the same buffer for 5 min at room temperature. An incubation of tetramethylbenzidine TMB-KPL substrate (Pierce) for 30 min at room temperature, followed by 1 N HCl was used for the colorimetric detection of titres endpoint. Optical densities at a wavelength of 450 nm (OD_{450nm}) were read using a microplate spectrophotometer (BioTek). All samples were tested in duplicate and the intra-assay variability was below 20%. Importantly, all protein solutions were centrifuged at 4 °C, 10 min, 8000×g to remove any precipitates prior use.

2.5. Circular dichroism spectroscopy

The recombinant (biotinylated and non-biotinylated) proteins were diluted in 300 mM NaCl, 50 mM Tris-HCl, pH 8.0 buffer (Buffer A) to the concentration of 100 µg/mL and dialyzed in Slide-A-Lyzer Dialysis Cassettes, 10K MWCO (Thermo Scientific). The dialysis procedure was performed during 12 h at 4 °C against 5 buffers containing decreasing concentrations of NaCl and Tris-HCl as follows: Buffer B – 225 mM NaCl, 50 mM Tris-HCl, pH 8.0; Buffer C –

150 mM NaCl, 40 mM Tris-HCl, pH 8.0; Buffer D – 75 mM NaCl, 30 mM Tris-HCl, pH 8.0; Buffer E – 45 mM NaCl, 25 mM Tris-HCl, pH 8.0; Buffer F – 30 mM NaCl, 20 mM Tris-HCl, pH 8.0. The dialyzed samples were concentrated using Vivaspin 6, 10,000 MWCO (Sartorius Stedim Biotech) and quantified by spectrophotometry. Circular dichroism data were collected on an Olis DSM17 Spectrometer. Scans were recorded in a 0.1 cm path length. Circular dichroism scans were carried out from 260 to 200 nm with 5 s averaging times, 1 nm step size, and 2 nm bandwidth at 25 °C. Spectra were corrected for a buffer blank and baseline molar ellipticity at 260 nm. Scan data were smoothed by the Stavitsky-Golay method.

2.6. Human peripheral blood mononuclear cells – PBMC

Study subjects included 1 HIV-negative control (ID: 4535) and 1 HIV-infected individual (ID: 30699). The HIV positive sample was obtained from an individual infected with clade B virus, and who were not under antiretroviral therapy. This individual exhibited, at the time of PBMCs sample collection, 14.5% of CD4⁺ cells, 75.1% of CB8⁺ cells and viral load of 259,144 copies/mL, as measured using Roche second generation assays. This participant was selected based on previous screening experiments (data not published), where this individual has shown to be among the ones exhibiting high antibody titers against the HIV-1 2F5 epitope. PBMCs, cryopreserved in fetal bovine serum – FBS with 10% dimethyl sulfoxide – DMSO and stored in liquid nitrogen (–160 °C), were thawed for immediate use. The PBMC samples were obtained from the Multicenter AIDS Cohort Study – MACS after approval by the Ethical Committee of the University of Pittsburgh.

2.7. Identification of memory B cell expressing anti-2F5 antibodies by flow cytometry

Patient-derived PBMCs were thawed and washed twice with 1X PBS, and then resuspended in fluorescence-activated cell sorter (FACS) buffer (PBS with 2% bovine serum albumin and 0.1% sodium azide). One million cells were used for surface staining with LIVE/DEAD Fixable Blue Dead Cell Stain kit (Invitrogen) at 4 °C for 20 min to exclude non-viable cells. Following live/dead cell staining, for the detection of surface anti-2F5 antibodies, the PBMCs were stained with fluorescently labeled monoclonal antibodies anti-CD19 (MHCD1917, Invitrogen), anti-CD27 (356413, BioLegend), anti-IgG (562025, BD Biosciences) and with the biotinylated versions of the Top7 (background control) or Top7-2F5 proteins (here termed Top7-Biotin and Top7-2F5-Biotin, respectively) at 4 °C for 1 h. Following, the PBMCs were incubated with fluorescently labeled streptavidin (BD Biosciences) at 4 °C for 20 min and fixed with 2% paraformaldehyde – PFA. After washing twice, cells were resuspended in 400 µL FACS buffer. IgG isotype controls and fluorescence minus one (FMO) negative controls were included for staining. Flow cytometry data was acquired using an LSR II cell analyzer (BD Biosciences). Non-viable cells were gated out of further analysis. Single lymphocytes were discriminated based on cell size and granularity using forward and side-scatter profiles. B cells were identified by surface expression of the CD19 marker (gated as CD19⁺ cells). Memory B cells were identified by surface expression of CD27 and IgG markers. While memory B cells expressing the anti-2F5 antibody were identified from their parent CD19⁺, CD27⁺ and IgG⁺ population by surface detection of binding to Top7-2F5-Biotin. Data were analyzed using FlowJo 887 software (FlowJo, LLC, Ashland, OR, USA).

3. Results and discussion

3.1. Affinity trend towards the human Anti-2F5 antibody

The first question to address the potential of the computationally designed protein Top7-2F5 to be used as a marker to identify specific B-cells was elucidated by measuring the affinity trend between the human anti-2F5 antibody and the Top7-2F5 protein by ELISA. The MPER peptide was also evaluated for comparison purposes, since it still remains as the gold standard antigen to detect anti-2F5 antibodies *in vitro*. The Top-2F5 protein and the MPER peptide were used as equal molarities to ensure proper comparison. The ELISA data showed that both molecules were recognized by the human anti-2F5 antibodies and, although the binding affinities have not been experimentally quantified, it is clear the superior performance of the recombinant protein when compared to the MPER (Fig. 1). The calculation of the half maximal effective concentration (EC₅₀) for both molecules revealed that the MPER peptide bound to the human anti-2F5 antibody with an EC₅₀ of 3.954 pM, while the Top7-2F5 protein exhibited an EC₅₀ value more than 100 times lower (0.037 pM). This data indicates that Top7-2F5 is recognized in a greater degree by the human anti-2F5 antibody when compared to the MPER and denotes that the designed protein is a more suitable antigen for future assays aiming *in vitro* detection of anti-2F5 antibodies than its counterpart.

3.2. Structural stability, conformational dynamics and electrostatic profile

The structural stability of Top7-2F5 and its biotinylated version (Top7-2F5-Biotin) at room temperature was experimentally assessed by circular dichroism (CD) spectroscopy, along with the native scaffold and its biotinylated counterpart (Top7 and Top7-Biotin, respectively). The far-UV CD spectra were collected from 200 to 260 nm, therefore allowing only for a qualitative comparison of secondary structure content. Nevertheless, the spectra were also used to evaluate overall folding and secondary structure content similarity between the devised protein and the original Top7 scaffold upon biotinylation. All four proteins displayed a CD

spectrum characteristic of α/β proteins (Fig. 2 A), which is in agreement with previously reported CD spectra for non-biotinylated Top7 [6,40]. The four spectra revealed two negative bands characteristic of α -helix content at 208 nm and 222 nm and a negative band correspondent to β -sheet content between 216 and 218 nm. Comparison between the spectra of Top7 and Top7-Biotin showed that the secondary structure content of both systems is comparable in magnitude (Fig. 2 A, upper graph). Similarly, no significant ellipticity changes at 208, 222 and 216–218 nm were observed when comparing Top7-2F5 and Top7-2F5-Biotin systems (Fig. 2 A, lower graph), suggesting a remarkable folding similarity between the two versions of each protein.

Subsequently, the molecular basis of the structural stability of Top7-2F5 protein upon biotinylation was computationally investigated by 100 ns molecular dynamics simulations. The structure of Top7 is known to be highly stable in a wide range of pH values (2–10) and temperatures (up to 90° Celsius). Aiming to assess the effect of adding an epitope and/or biotinylation on protein thermal stability, simulations at three different temperatures (300 K, 340 K and 380 K) of the modified proteins were carried out and compared to the native Top7. The average root-mean-square deviation (RMSD) for position of all backbone atoms was calculated over the last 50 ns of each trajectory. The data agreed with previous studies [7,41] showing that the Top7 and Top-2F5 systems kept their native-like structure at 300 K, with an average RMSD of ca. 0.19 ± 0.04 nm and 0.24 ± 0.02 nm, respectively (Fig. 2 B). As the simulation temperature increased, the averaged RMSD assumed higher values: 0.35 ± 0.04 nm and 0.47 ± 0.07 nm at 340 K and 380 K, respectively, for Top7 (Fig. 2 B, upper graph) and 0.47 ± 0.04 nm and 0.50 ± 0.06 nm at 340 K and 380 K, respectively, for Top7-2F5 (Fig. 2 B, lower graph). This is expected due to effects on the amount of thermal energy driving unfolding [42]. Therefore, the higher RMSD values suggest that the studied proteins undergo substantial conformational changes and do not keep the overall folding of the native structure at higher temperatures. On the other hand, the biotinylated proteins consistently showed lower RMSD values, regardless the simulation temperature. The low RMSD values suggest that both biotinylated proteins remain as native-like even at relatively high temperatures, pointing towards the higher stability of the biotinylated systems in comparison to their non-biotinylated forms.

The average positional root-mean-square fluctuation (RMSF) for the backbone atoms was also calculated for all systems over the last 50 ns (Fig. 3). For the native Top7 protein, the regions of highest atomic fluctuations in all simulated temperatures corresponded to regions of turn/loops and α -helix motifs (Fig. 3 A). The RMSF values increased as the temperature increased, with the average fluctuations in the α -helix regions at 380 K reaching values three times higher than what is observed at 300 K. On the other hand, the RMSF of Top7-Biotin remained low (below 0.2 nm) throughout the simulation time for 300 K and 340 K temperature simulation. For the simulation at 380 K the RMSF increases up to 0.35 nm, indicating a structural disruption in the first α -helix. Similarly, for the Top7-2F5 protein (Fig. 3 B), the highest RMSF fluctuations corresponded to regions of turn/loops and α -helix motifs, including the 2F5 epitope loop. As expected, the RMSF values increased following the temperature increase and were notably high in the epitope region at 380 K, where the RMSF reached values three times higher than what was observed at 300 K. The analysis of the positional fluctuations of Top7-2F5-Biotin revealed that the RMSF values at 300 K were overall slightly lower when compared to the non-biotinylated Top7-2F5. Interestingly, the 2F5 epitope loop becomes even more flexible within the Top7-2F5-Biotin protein. This is likely due to the biotinylation onto the K25 residue located in the middle of the epitope, which may represent a barrier for antibody

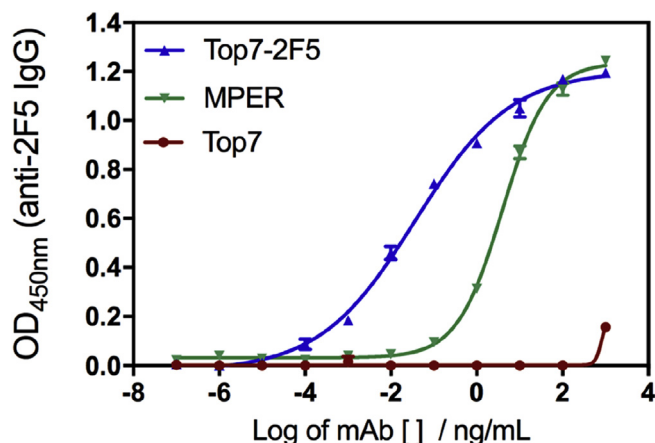


Fig. 1. Affinity trend measurement of the human anti-2F5 antibody against the Top7-2F5 protein and MPER peptide by ELISA. The Top7-2F5 protein and the MPER peptide were immobilized on the surface of ELISA plates at equal molarities (32 pmol) and their immunoreactivity against a serial dilution of the 2F5 mAb, ranging from 1 ng mL^{-1} (6.67 pM) to $1 \times 10^{-13} \text{ ng mL}^{-1}$ ($6.67 \times 10^{-13} \text{ pM}$) was assessed. The protein scaffold (Top7) was used as background control and showed no unspecific binding to the 2F5 mAb. Each line represents a single mAb curve measured in duplicates. The curves are color-coded as follows: Top7-2F5 in blue, MPER in green and Top7 in red.

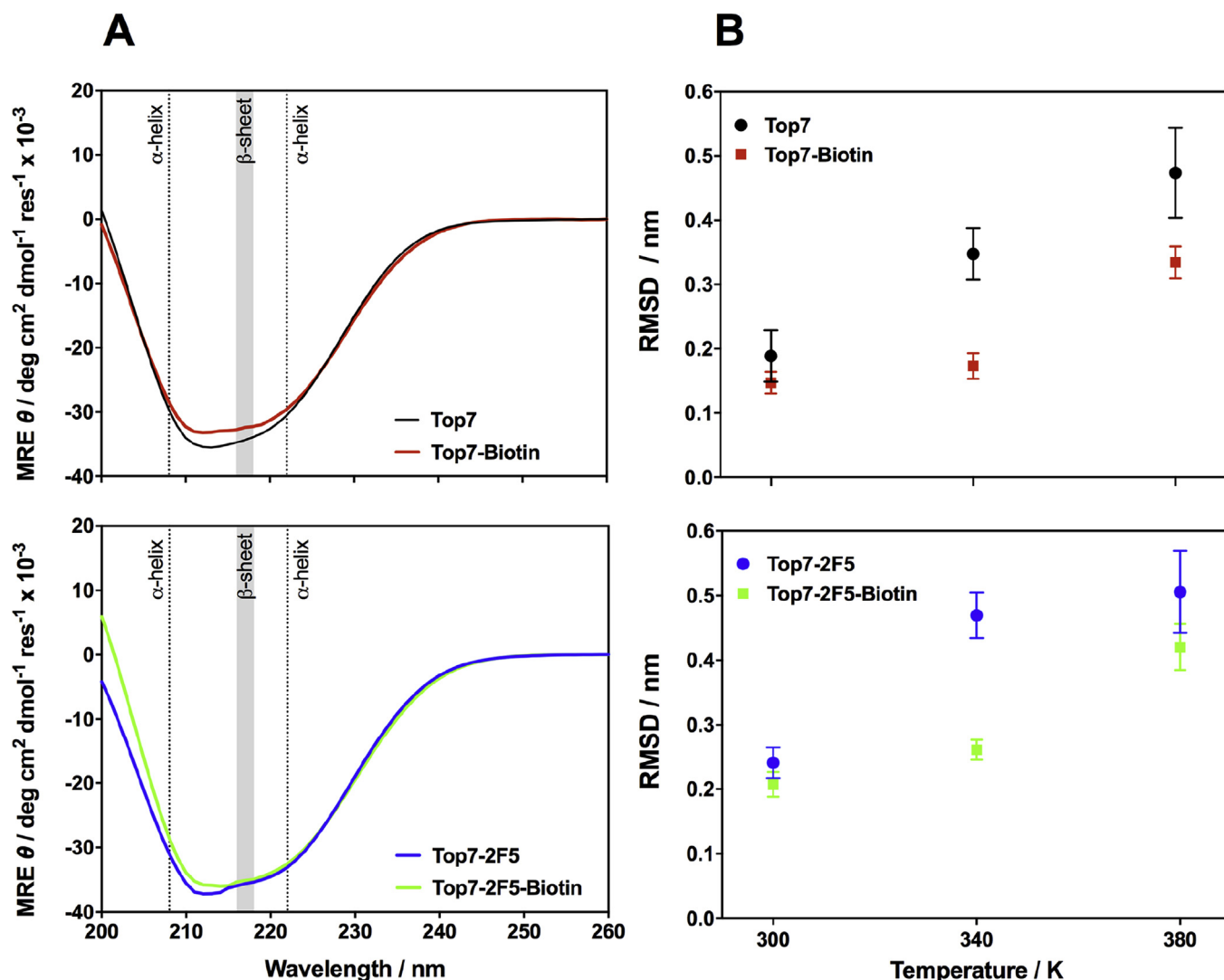


Fig. 2. Structural characterisation of the recombinant Top7 and Top7-2F5 proteins. (a) Far-UV circular dichroism spectra of the Top7 protein (upper graph) and Top7-2F5 (bottom graph) in their native and biotinylated versions collected at 25 °C. Dashed lines mark 208 nm and 222 nm, points in the spectrum corresponding to helical content, while the region from 216 to 218 nm corresponding to β -sheet content is highlighted by a shaded band. MRE stands for mean residue ellipticity. (c) Average root mean square deviation (RMSD) for the last 50 ns of each MD simulation trajectory for all simulated systems: Top7 (black), Top7-Biotin (red), Top7-2F5 (blue) and Top7-2F5-Biotin (green). Results indicate that biotinylation process confers structural stability to the proteins. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

recognition. In agreement with what was observed for Top7 and its biotinylated form, the RMSF values for Top7-2F5-Biotin remained low (below 0.25 nm) throughout the simulation time, regardless the temperature. This finding also suggests that biotinylation of Top7 and Top7-2F5 confers increased structural stability to the devised proteins.

As expected from the RMSD and RMSF data, the time-dependent secondary structure profiles for the four analyzed systems showed that all secondary motifs were well maintained over the entire trajectories at 300 K (Fig. 3C). For the Top7 protein, minor disturbances in the helical content were observed at 340 K. However, as the temperature increased to 380 K, a more dramatic change in secondary structure is observed, with a permanent destabilization of both α -helices, along with the increase of β -sheet content, indicating unfolding of this protein. In contrast, the time-dependent secondary structure profile of Top7-Biotin does not show any partial nor permanent disruption of secondary structure at 300 K and 340 K temperatures. Disruption only occurs in the first

α -helix at 380 K, suggesting that biotinylation provides increased structural stability. A similar behavior was observed for Top7-2F5, where the non-biotinylated form exhibited remarkable and progressive destabilization of the first α -helix as temperature increases and the appearance of β -sheet content on the epitope loop region. On the other hand, Top7-2F5-Biotin follows the same trend observed for Top7-Biotin, without any apparent disruption of secondary structure content at the analyzed temperatures. Loss of helical content is only observed at the highest temperature (380 K). It is known that the β -sheet core of Top7 is responsible for its remarkable stability and that its α -helices are not as stable.⁸ During the simulations, it was observed that biotin molecules contribute to lower the flexibility of the α -helices by making a hydrogen bond network that “locks” the two helices (Fig. S3), therefore conferring increased stability to the biotinylated forms.

A comparison of the backbone of the 2F5 epitope within Top7-2F5 with its crystallographic structure as recognized by the respective mAb (PDB ID: 1TJI) [2] indicates that the grafted epitope

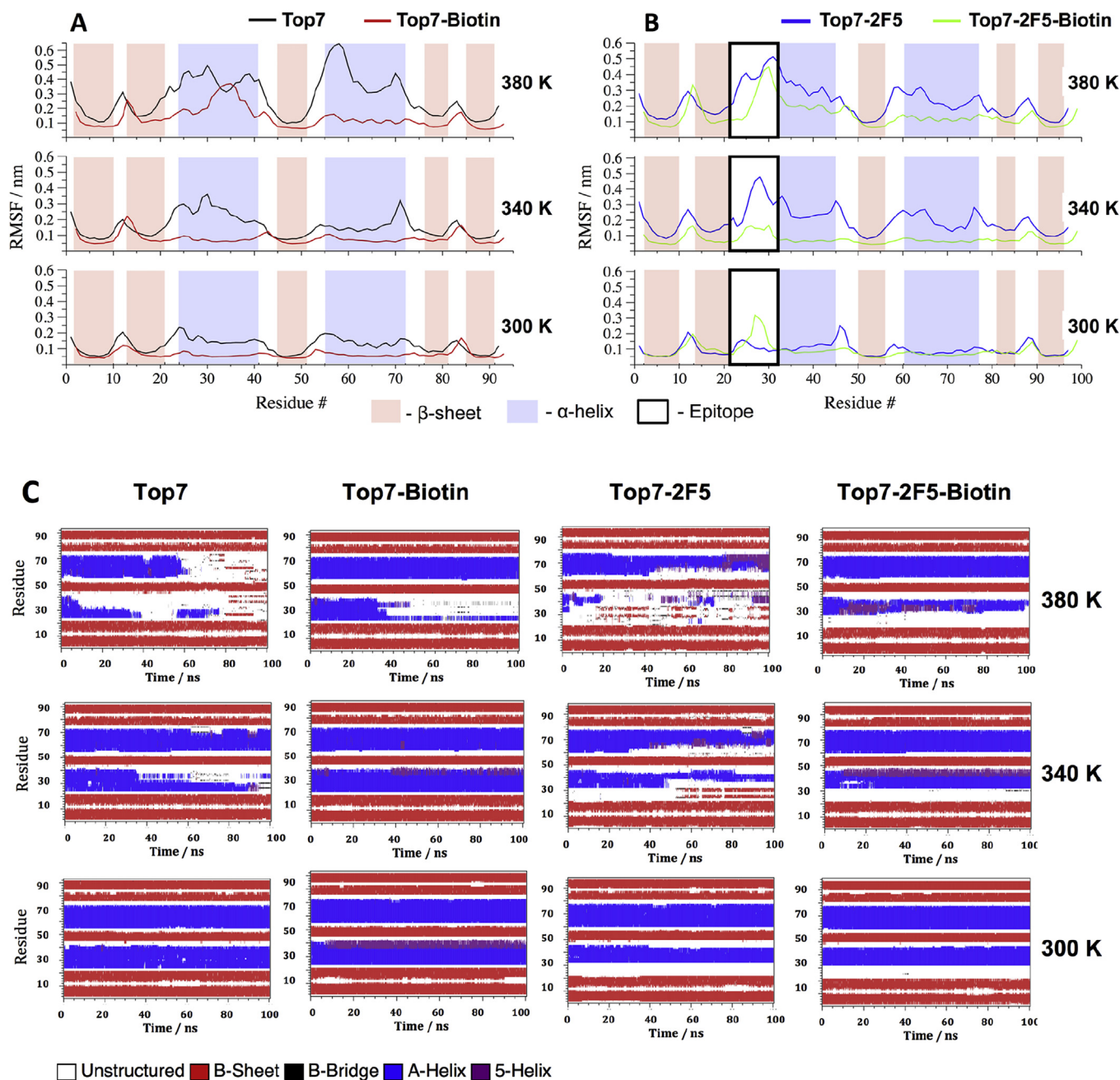


Fig. 3. Positional quadratic mean fluctuation (RMSF) for the backbone atoms and Time-dependent secondary structure profiles (DSSP) analysis. (a) RMSF values for the Top7 (black line) and Top7-Biotin (red line), indicating that regions of greatest atomic fluctuations correspond to regions of turn/loops and α -helix motifs. The RMSF values for Top7 increase with increasing temperature and for Top7-Biotin remained with lower values regardless of temperature. (b) RMSF values for the Top7-2F5 (blue line) and Top7-2F5-Biotin (green line) showing that with RMSF values increase with increasing temperature and for Top7-2F5-biotin the RMSF values were overall slightly lower when compared to Top7-2F5. The 2F5 epitope loop becomes more flexible in the Top7-2F5-Biotin protein due to a biotinylation at the 25LYS residue located in the middle of the epitope. (c) DSSP for all four systems, indicating that all secondary structures were well maintained in all trajectories at 300 K. As the simulation temperature increases, native proteins (Top7 and Top7-2F5) lose structural content. In contrast, the DSSP profile of the biotinylated proteins undergoes fewer perturbations with increasing temperature, indicating that biotin confers structural stability. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

exhibits one major conformation (Fig. 4 A). In contrast, in the biotinylated protein, the epitope is more flexible and displays two major populations (Fig. 4 B). As electrostatic properties are known to be important for antibody-reactivity and specificity [43], the electrostatic surface potentials were also calculated and plotted onto the molecular surface of representative frames for Top7-2F5 and Top7-2F5-Biotin (Fig. 4C and D, respectively). A significant divergence can be observed as Top7-2F5-Biotin shows a remarkably overall negatively charged profile. It is expected, as the biotin

cancels out positively charged solvent-exposed amines. Top7-2F5 displays a more homogeneous positive and negative charged regions distribution. Remarkably, surface charge changes in the epitope region are subtler (Fig. 4C and D, areas highlighted in green). Taken into account dynamics and the electrostatic profile of the epitope, the findings suggest that biotinylation may impact the interactions (specificity and/or affinity) between the epitope and the mAb, but it should still be recognized. On the other hand, an increase in negative net charge is more likely to improve the

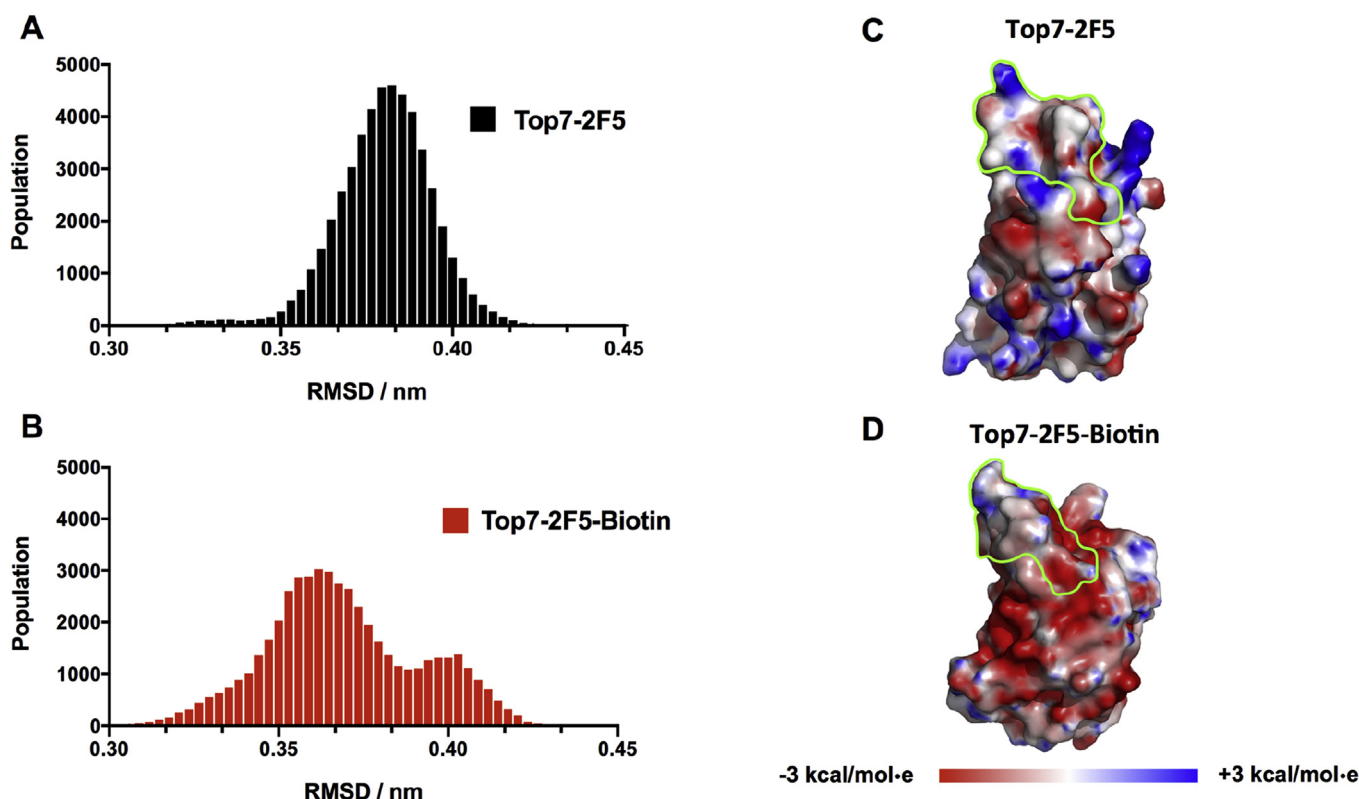


Fig. 4. Epitope's RMSD distribution and electrostatic potential analyses for Top7-2F5 and Top7-2F5-Biotin proteins. Population distribution of RMSD values between the epitope's backbone atoms in protein (a) Top7-2F5 and (b) Top7-2F5-Biotin. Values are relative to the conformation of the epitope when complexed to antibody mAb 2F5 (PDB 1TJ1), indicating that the non-biotinylated epitope has a major preferential conformation, whereas the biotinylated one has two different conformations. (c) Electrostatic potential surface of the proteins Top7-2F5 and (d) Top7-2F5-Biotin. The location of the epitope is highlighted in green. Top7-2F5-Biotin shows a remarkably more negative profile, while Top7-2F5 exhibits a more homogeneous positive and negative regions distribution. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

solubility by preventing aggregation and precipitation of the biotinylated proteins [44,45].

In summary, according to both techniques, at room temperature, all proteins display similar overall structure. In all cases, the shape of the CD spectrum is conserved suggesting retention of native structure, corroborating with the computational predictions that indicates biotinylation of the proteins does not compromise their folding. Neither the insertion of the 2F5 epitope onto Top7, nor the biotinylation of both proteins seem to cause significant changes in the Top7 scaffold or the secondary structure content at room temperature. Nevertheless, the 2F5 epitope displays a higher flexibility on the biotinylated Top7-2F5 protein. It is likely due to the presence of a biotin residue right in the middle of the epitope sequence. On the other hand, the simulation results show that biotin stabilizes thermal unfolding by retaining secondary structure content as temperature rises.

3.3. Assessing 2F5 mAb binding upon Top-2F5 biotinylation

In order to ascertain that biotinylated Top7-2F5-Biotin maintained its immunoreactivity, protein recognition by the 2F5 monoclonal antibody was evaluated by means of ELISA. For these assays, Top7-2F5 and Top7-2F5-Biotin were bound to the solid phase at equal molarities and streptavidin was used to confirm the biotinylation reaction, while the 2F5 mAb was used to assess overall immunoreactivity (Fig. S4). Long-term stability of both proteins was also assessed at several time points during the period of 18 months (Fig. S5). Fig. 5 shows similar immunoreactivity for both proteins when freshly prepared. The slightly lower reading for

Top7-2F5-Biotin may be attributed to the increased flexibility of the epitope region and differences in the electrostatic surface potential in the epitope region, as showed in the simulations. On the other hand, immunoreactivity for the biotinylated protein decreases significantly as a function of storage time, whereas it remains unchanged by the non-biotinylated protein throughout the experiment time period. The results show that antibody detection by Top7-2F5-Biotin can be safely measured only if the protein has been produced and properly stored within six months. It is worth noting that all ELISAs were performed using the same protein concentration from the soluble fraction, in all time points for both proteins. Therefore, the reason for the decreased affinity of Top7-2F5-Biotin as a function of time remains unclear at this point.

Interestingly, while the non-biotinylated protein aggregates over time, the biotin-bound protein did not show aggregation as a function of time (Fig. S5). This is likely due to the increase in protein solubility upon biotinylation, which acts preventing aggregation and precipitation, as well as providing structural stabilization. Support for the former is provided by comparing the electrostatic surface profile of the biotinylated and non-biotinylated proteins (Fig. 4C). Evidence for structural stabilization is also offered by the simulations. While the effects of temperature and time are not directly compared, Fig. 3 B shows that even a modest increase in temperature shows loss of helical content and increase of sheet content for the non-biotinylated Top7-2F5 protein, a typical effect prior protein aggregation. This does not occur when the recombinant protein is biotinylated for temperatures up to 380 K, for the same simulation period.

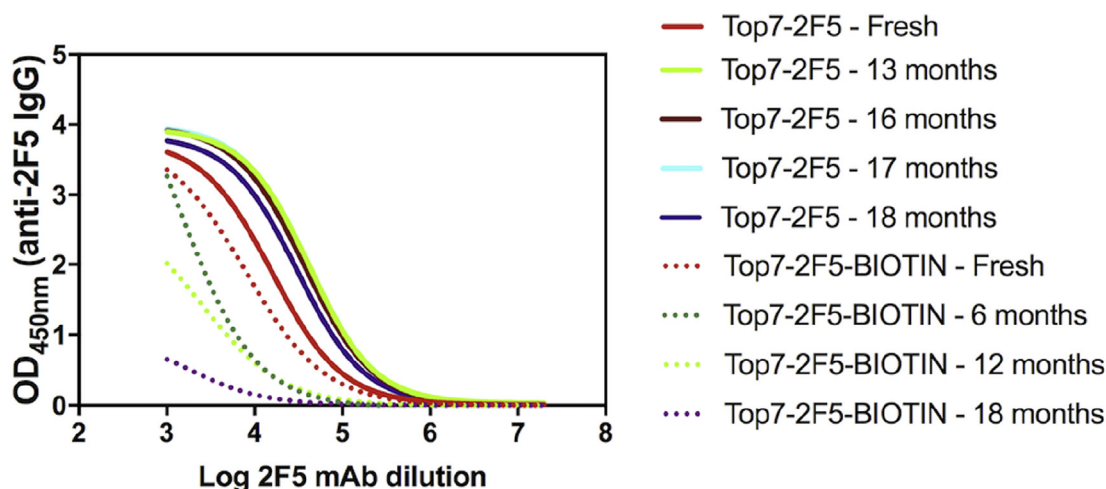


Fig. 5. Kinetics of 2F5 antibody binding to the native and biotinylated versions of Top7-2F5 protein as measured by ELISA. The Top7-2F5 native and biotinylated proteins were immobilized on the surface of ELISA plates and their immunoreactivity over time was assessed against a serial dilution of the 2F5 mAb ranging from 1:1,000 to 1:19,683,000. Each line represents a single 2F5 mAb curve. All measures were performed in duplicates.

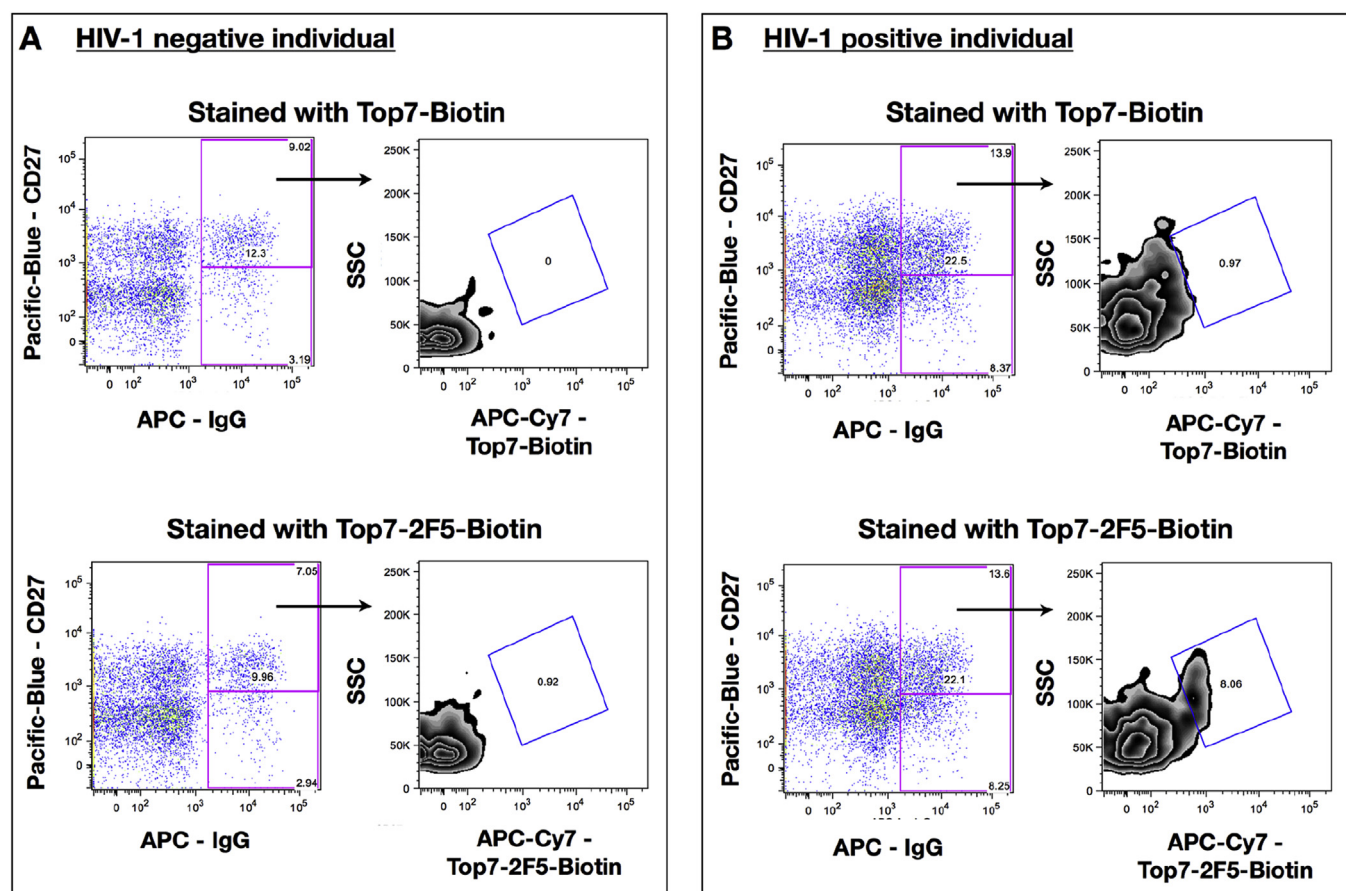


Fig. 6. Detection of surface expression of anti-2F5 antibodies by human B-cells through flow cytometry. PBMCs from a healthy individual (a) and from an HIV-1 positive individual (b) were stained with anti-CD19, CD27, IgG, along with the biotinylated Top7 (upper graphs) and Top7-2F5 proteins (bottom graphs) for evaluation by flow cytometry. Dead cells were excluded by LIVE-DEAD staining, and the blood lymphocyte population was gated to determine the percentages of classical memory B cell subsets, as defined by co-expression of the CD19, CD27 and IgG markers (highlighted in pink boxes). The percentage of B cells expressing anti-2F5 antibodies from each human sample was determined within the classical memory B cell population by measuring the fluorescence of the biotinylated Top7-2F5 protein. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.4. Detection of B-cells displaying the 2F5 antibody

Once the immunoreactivity of the biotinylated Top7-2F5 protein was confirmed, its potential use as a marker to identify human B cells expressing 2F5 antibodies was assessed through flow cytometry assays. As a proof of principle, patient-derived PBMCs from one HIV-negative and one HIV-positive individual were stained for the presence of the 2F5 antibodies on the surface of B-cells (CD19⁺, CD27⁺ and IgG anti-2F5⁺). Our data show that the devised biotinylated protein was recognized by B-cells from an HIV-1 positive individual expressing the target antibody (Fig. 6). It is worth mentioning that a sample from only one HIV-1 positive individual was used to demonstrate the validity of the devised protein to be recognized by patient-derived PBMCs expressing anti-2F5 antibodies. However, anti-HIV-1 2F5 antibodies are only produced by a minority of HIV-infected individuals [3], and this HIV-positive sample was previously screened for the presence of anti-2F5 antibodies in plasma (data not published), which suffices its use in the proof of principle assay. No expressive unspecific binding was observed when PBMCs from an HIV-1 negative individual were stained with the Top7-2F5-Biotin. Accordingly, no unspecific binding to the Top7 scaffold was observed. Together, these data confirms the potential of using the biotinylated form of Top7-2F5 protein to identify 2F5 epitope-specific B-cells and open the venue for future experimental studies broadening the spectrum of clinical samples and HIV-1 patient groups (e.g., HIV-1 progressors and non-progressors), and allowing for assessment of the immunoreactivity and protective potential of vaccine antigens *in vitro*.

4. Conclusions

The work presented here demonstrates that the Top7-2F5 is a Top7 native-like protein with higher affinity to human anti-2F5 antibodies when compared to MPER, the actual gold standard used to detect such antibodies. Upon biotinylation, the protein displays increased stability and solubility and it can be stored for several weeks before losing its immunoreactivity. The biotinylated protein Top7-2F5 strongly reacts with its corresponding neutralizing antibody *in vitro* and can be recognized by anti-2F5 antibodies produced by B-cells from HIV-1-infected individuals. Together, our data show that Top7-2F5-Biotin can be used as a marker to identify B-cells producing anti-2F5 antibodies. Although preliminary, this finding lays the foundation to use this protein to quantify the success and protective effect of prophylactic and therapeutic interventions in the field of HIV vaccines by identifying B-cells producing anti-2F5 antibodies in response to immunization strategies based on the HIV envelope glycoprotein.

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Ethical approval

The human PBMC samples used in this study were provided by the MACS cohort group after approval by the Ethical Committee of the University of Pittsburgh.

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Appendix A. Supplementary data

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